

Accuracy and Explainability in Clinical Machine Learning

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Abstract

Modern ensemble models such as Random Forests achieve strong predictive accuracy but offer no explanation for their decisions a critical limitation in domains like healthcare where transparency and accountability are essential. This project addresses the interpretability problem by implementing a post-hoc rule extraction pipeline that automatically distills a trained Random Forest into a compact set of human-readable IF-THEN logical rules. We apply this to the Breast Cancer Wisconsin dataset, a clinically meaningful collection of tumor biopsy measurements where interpretability has real-world significance. Our approach trains a Random Forest as a baseline for comparison of the accuracy, it extracts candidate decision rules by traversing the internal structure of every tree, and uses Lasso regularization to identify and retain only the rules that carry genuine predictive signal discarding redundant and noisy ones. We aim to analyze the tradeoff between interpretability and accuracy, examine what the surviving rules reveal about the structure of the data, and evaluate whether the extracted rules reflect clinically meaningful patterns.

1 Introduction

Machine learning models have demonstrated remarkable predictive power across a wide range of real-world applications. In the medical domain particularly, models trained on clinical data have shown the ability to match or exceed human expert performance on diagnostic tasks. However, the models that tend to perform best are ensemble methods like Random Forests, are also the hardest to interpret. No single human can read 100 decision trees and extract a coherent explanation for why a specific patient was classified as malignant or benign. This opacity is not merely an inconvenience it is a fundamental barrier to deployment in high-stakes settings where doctors, patients, and

regulators require explicit justification for every decision.

To resolve this we propose a three-stage strategy:

1. **Random-Forest Baseline:** We train a Random Forest on the Breast Cancer Wisconsin dataset to establish a strong, accurate uninterpretable baseline. This gives us the performance ceiling we aim to approximate with an interpretable model.
2. **Rule Extraction:** We implement a recursive tree-traversal algorithm that exhaustively walks every tree in the forest and collects every root-to-leaf path as a human-readable IF-THEN logical rule. A forest of 100 trees produces thousands of such candidate rules, most of which are redundant or noisy.
3. **Sparse Rule Selection via Lasso:** We formulate rule selection as a sparse linear regression problem. A binary indicator matrix encodes which rules apply to which training instances, and L1 regularization, Lasso is applied to automatically zero out redundant and uninformative rules. The surviving non-zero rules form our final interpretable rule set.

2 Methodology

Our methodology will consist of four main components: dataset preparation, base model training, rule extraction, and sparse rule selection via Lasso regularization.

2.1 Dataset and Preprocessing

We began our experiments with the Breast Cancer Wisconsin dataset as it provides a clean, well-structured binary classification task with a direct and unambiguous outcome a tumor is either malignant or benign. This made it the ideal starting point to validate that our pipeline works correctly before applying it to more complex settings. Once

we have established the approach on this dataset, we plan to extend our experiments to the Pima Indians Diabetes dataset, where the boundary between disease and non-disease states is influenced by a combination of interacting factors such as glucose levels, BMI, age, and insulin making it a more challenging and interesting test of whether our rule extraction pipeline generalizes beyond clean binary problems.

We use the Breast Cancer Wisconsin (Diagnostic) dataset, available directly through scikit-learn, which contains 569 patient records each described by 30 numerical features extracted from fine needle aspirate biopsy images. The target variable is binary malignant (212 patients) or benign (357 patients) and we split the data 80/20 into training and test sets using stratified sampling.

2.2 Base Model: Random Forest

To establish a baseline, we trained a Random Forest of 100 decision trees, each with a maximum depth of 5, on the training split. This gives us a strong, accurate reference model whose internal structure we will attempt to distill in the subsequent steps. This model would be our reference point. We ran it on the held-out test set and recorded accuracy, F1 score, and a full classification report so we have a concrete performance ceiling to measure against later. We also pulled the Gini feature importances out of the trained forest these will later serve as a reference point for validating whether our extracted rules identify the same signals the forest itself considered important.

2.3 Rule Extraction: Tree Traversal

Once the Random Forest is trained, we plan to implement a recursive tree traversal algorithm that will go through every tree in the forest from root to leaf and collect every complete decision path as a human-readable IF-THEN logical rule. Each path through a tree of depth 5 represents a conjunction of up to 5 conditions for example, IF `worst_area > 880 AND worst_concave_points > 0.13 AND mean_texture > 21` THEN Malignant. A forest of 100 trees is expected to produce thousands of such candidate rules. We intend to apply a purity filter to keep only rules whose leaf nodes are strongly dominated by one class, removing uncertain or mixed predictions from the candidate pool.

2.4 Binary Indicator Matrix Construction

To prepare the candidate rules for Lasso optimization we plan to construct a binary indicator matrix A of dimensions $N \times K$, where N is the number of training instances and K is the number of candidate rules. An entry $A_{i,k}$ will equal 1 if training instance i satisfies all conditions of rule k , and 0 otherwise. This matrix will capture in mathematical form the relationship between every patient and every candidate rule, and will serve as the direct input to the Lasso optimization step.

2.5 Sparse Rule Selection: Lasso

We plan to formulate rule selection as a sparse linear regression problem. Given the binary indicator matrix A and the true class labels y , we intend to solve the following optimization objective:

$$\min_{\alpha} \frac{1}{2N} \|y - A\alpha\|_2^2 + \lambda \|\alpha\|_1 \quad (1)$$

The L_1 penalty should force the coefficients of redundant, noisy, or highly correlated rules to exactly zero. Only rules with non-zero coefficients after optimization will survive as our final sparse rule set. We plan to use LassoCV with 5-fold cross validation to automatically determine the optimal regularization strength λ . We also intend to experiment with varying λ across a range of multipliers to study the tradeoff between the number of surviving rules and predictive accuracy. We plan to formulate rule selection as a sparse linear regression problem, following the rule ensemble framework proposed by Friedman and Popescu (1).

2.6 Conflict Resolution methodology

When multiple rules from the sparse rule set fire for a single patient but disagree on the predicted class, a conflict resolution mechanism will be needed. We plan to resolve conflicts using a weighted vote the side whose surviving rules have higher total absolute Lasso weight will win. This should ensure that stronger, more confident rules override weaker ones rather than relying on a simple majority count.

2.6.1 SIRUS: Stable and Interpretable Rule Set for Classification (2)

We also will try and implement a stable rule extraction algorithm to ensure our identified patterns remain consistent even if the underlying dataset changes slightly. The process starts by optionally preprocessing the dataset, so all data input falls into discretized bins representing the full range of

the dataset for that feature. We then, as before construct a large random forest from the training dataset. We restrict the depth of these trees to keep the individual rules short and understandable. We then extract every path from the top of the tree down to the leaves across the entire forest. Instead of using a linear penalty to filter these candidate rules, we count how frequently each rule appears across all the generated trees. A rule that appears often is highly robust and not merely a product of random noise. We select only the rules whose frequency of occurrence exceeds a specific threshold. These highly frequent rules form our final rule set. This approach guarantees that the resulting logical statements are not only accurate but highly stable, meaning we would extract the exact same explanation even if we trained the model on a slightly different sample of patients.

3 Experiments and Evaluation

3.1 Dataset Choice and Motivation

We chose the Breast Cancer Wisconsin (Diagnostic) dataset as our primary experimental dataset because the classification task is binary and the outcome is direct and unambiguous a tumor is either malignant or benign, making it the ideal starting point to validate our pipeline before moving to more complex settings.

Next we plan to try and implement our pipeline on the Pima Indians Diabetes dataset as a second experimental setting. Diabetes is a more complex classification problem because the outcome is influenced by a combination of factors glucose levels, BMI, age, insulin, and blood pressure that interact with each other in non-trivial ways, unlike breast cancer where the diagnosis is driven by direct physical measurements of the tumor. This makes it a more challenging and interesting test of whether our rule extraction pipeline can still produce interpretable and accurate rules when the underlying structure of the data is less clean. We plan to run these experiments as the next step in our project once we have fully validated our approach on the breast cancer dataset.

3.2 Train-Test Split

We split the Breast Cancer Wisconsin dataset into 80% training and 20% test, giving 455 training instances and 114 test instances. We used stratified sampling to ensure that the proportion of malignant and benign cases is preserved in both splits,

without stratification, a random split could produce a test set that is unrepresentative of the true class distribution, which would make our evaluation metrics unreliable. The test set is held out entirely and is not used at any point during model training, rule extraction, or Lasso optimization. It is only used at the very end to evaluate the final model, ensuring our reported numbers reflects genuine out-of-sample performance.

3.3 Evaluation Metrics and Success Criteria

We will evaluate our system across three dimensions,

1. **Predictive Fidelity:** We will measure accuracy and F1 score for both the base Random Forest and the final sparse rule set on the held-out test set. How much accuracy are we willing to trade-off for interpretability? We will try to minimise this and if possible to keep it same as the baseline model.
2. **Cognitive Interpretability:** We measure the total number of rules retained after Lasso and the average number of conditions per rule. Our target is a compact set that a domain expert could realistically read and reason about. We will also examine whether the surviving rules are logically coherent do the conditions within each rule make sense together?
3. **Structural Feature Alignment:** We measure the overlap between the features that appear in our sparse rules and the top features identified by the Random Forest’s own Gini importance rankings. High overlap means our extraction process identified the same signals the baseline model (random forest) found important validating that we captured real structure rather than noise. Low overlap would suggest our rules are picking up on artifacts rather than genuine patterns in the data.

At this stage we have completed the dataset preparation, base model training, and have begun exploring the rule extraction pipeline. The results so far are encouraging and suggest the approach is viable. Over the coming weeks we plan to complete the full Lasso-based rule selection, conduct the evaluation across all three dimensions described above, and extend the experiments to the Pima Indians Diabetes dataset.

References

- [1] Friedman, J. H., & Popescu, B. E. (2008). *Predictive learning via rule ensembles*. The Annals of Applied Statistics, Vol. 2, No. 3, 916–954. <https://doi.org/10.1214/07-AOAS148>
- [2] Bénard, C., Biau, G., da Veiga, S., & Scornet, E. (2021). *SIRUS: Stable and Interpretable RULe Set for Classification*. arXiv:1908.06852 [stat.ML]. <https://doi.org/10.48550/arXiv.1908.06852>